AKASH VIJAYAKUMAR

MSc Robotics with Industrial Application, Heriot-Watt University, Edinburgh

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PERSONAL PROFILE

MSc Robotics with Industrial Applications student at Heriot-Watt University, Edinburgh, with a strong background and hands-on experience in robotics, autonomous navigation, collision avoidance, docking algorithms particularly in autonomous surface vessel (ASV) and underwater (AUV) systems. Experienced in Reinforcement Learning (PPO, SAC, DQN), Inverse RL, SLAM, Model Predictive Control (MPC), Model Predictive Path Integral Control (MPPI) and multidisciplinary team environments through research at a top global institute, IIT Madras and through international internships. Qualified second toughest examination in the world (IIT JEE Advanced). Passionate about translating software into real world physics, with hands-on experience in sensor integration to robotic systems. Five peer-reviewed publications in international conferences on autonomous surface vessel control and collision avoidance and two international competition wins in marine autonomous navigation challenges. Experienced in building robotic simulations using pybullet engine, MuJoCo and 3JS framework. Currently working on developing a Multi Agent RL framework for AUV navigation, collision avoidance and manipulator control in Ocean Systems Lab, HWU, Edinburgh.

EDUCATION

MSc Robotics with Industrial Application — *2026 – 2028*
Heriot-Watt University — *Edinburgh, UK*
Semester 1: Grade A

BTech, Naval Architecture and Ocean Engineering — *2021 – 2025*
Indian Institute of Technology (IIT) Madras — *Chennai, India*
CGPA: 7.90/10

Class XII (Higher Secondary) — *2020*
GVHSS Alamcode — *Kerala, India*
Score: 98.5% (100/100 in Physics, Chemistry and Mathematics)

Class X — *2018*
GHSS Navaikulam — *Kerala, India*
Score: 10/10 GPA

ACADEMIC ACHIEVEMENTS

- Selected as Student Poster Competition participant among 21 international students chosen from 230 applicants at **IEEE Oceans Halifax 2024, Canada**.
- Secured **All India Rank 7,135** in JEE Advanced (2021) among 250,000 candidates (World's Second Toughest Examination).
- Secured **All India Rank 9,650** in JEE Mains (2021) among 1.5 million candidates.
- Secured **All India Rank 27** in Kerala Engineering Entrance Examination among 50,000 candidates.

RESEARCH PUBLICATIONS

Extraction of Ship Collision Avoidance Patterns from AIS Data
OMAE 2024, Singapore (Presented & Published) | Akash Vijayakumar, Amar Nath Singh, Abhilash Somayajula

Deep Reinforcement Learning for Ship Collision Avoidance and Path Tracking
OMAE 2024, Singapore (Presented & Published) | Amar Nath Singh, Akash Vijayakumar, Shankruth

Balasubramaniyam, Abhilash Somayajula

Data Driven Identification and Model Predictive Control for a Catamaran Surface Vessel
OMAE 2024, Singapore (Presented & Published) | Vallabh Deogaonkar, Mohd. Ibrahim, Akash Vijayakumar, Abhilash Somayajula

Learning Autonomous Docking Operation of Fully Actuated Autonomous Surface Vessel from Expert Data
IEEE Oceans 2024, Halifax (Presented & Published) | Akash Vijayakumar, M A Atmanand, Abhilash Somayajula

Model Predictive Path Integral Docking of Fully Actuated Surface Vessel
Underwater Technology 2025, Taiwan (Presented & Published) | Akash Vijayakumar, MA Atmanand, Abhilash Somayajula

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, MATLAB, Arduino

Robotics & Autonomy: ROS, ROS 2, Micro ROS, Gazebo, MuJoCo, Reinforcement Learning (RL), PID, Imitation Learning, Inverse Reinforcement Learning, Model Predictive Control (MPC), MPPI, NeRFs, UWB, RTK, ESP, Embedded Systems, Sensor Fusion, Extended Kalman Filter (EKF), Motion Planning

Libraries & Frameworks: PyTorch, OpenCV, Pandas, NumPy, Simulink, Sympy, PCL, Open3D, pyserial, YOLOv8, Stable Baselines, Ray, PyBullet

Design & Tools: Fusion 360, CAD, Docker, Git (Version Control), Canva, Figma

Practices: Object-Oriented Programming

Relevant Courses: Manoeuvring and Control of Marine Vehicles, Introduction to Motion Planning, Advances in Machine Learning for Engineering Problems, Instrumentation for Ocean Technology, Data Structures and Algorithms (Python), Linear Algebra for Engineers, Differential Equations, Controls of Automotive Systems, Introduction to Field and Services Robotics, Machine Learning for Ocean Engineers

PROFESSIONAL EXPERIENCE

Researcher – Multi Agent Reinforcement Learning for Underwater Vehicles *May 2026 – Present*
Ocean Systems Lab, Heriot Watt University *Edinburgh, Scotland, UK*

- Working on Reinforcement learning based methods for Multi AUV navigation, collision avoidance and underwater manipulator control.

Research Intern – Acoustic Channel Communications Simulation *May 2025 – Aug 2025*
INESC TEC, University of Porto *Porto, Portugal*

- Developed acoustic channel communication simulation modelling under Dr Nuno Alexandre Cruz at the Institute for Systems and Computer Engineering, Technology and Science (INESC TEC).

Research Guild Student – MAVLAB, IIT Madras *Sep 2022 – May 2025*
Marine Autonomous Vehicle Laboratory (MAVLAB) *IIT Madras, India*

- Researched model predictive path-integral control and docking operation of autonomous surface vessels under Dr Abhilash Somayajula.
- Developed in-house UWB sensor-based indoor localisation for wave-basin testing in GPS-denied environments.
- Represented IIT Madras and presented a paper on "Extraction of collision avoidance patterns from AIS data" at OMAE 2024 Conference in Singapore.
- Selected and presented as a student poster competition representative at **IEEE Oceans Halifax, Canada** for research on "Learning Autonomous Docking Operation of Fully Actuated ASV using Expert Data".
- Developed inverse reinforcement learning-based solution for autonomous docking operation planning, from perception for a fully actuated catamaran vessel.
- Developed model predictive predictive control solution for docking of a fully actuated autonomous surface vessel using Gaussian Process regression techniques from free running data.
- Developed 6-DOF surface vessel dynamic model based reinforcement learning environment in Ray architecture.
- Developing swarm algorithm of ASVs and architecture for establishing cooperative autonomy between ASVs.
- Worked on the development team of custom 6-DOF in-house simulation model (MAVSIM) of ASVs using Manoeuvring Modelling Group (MMG) model of KVLCC2 oil tanker model for the lab.
- Developed 6-DOF surface vessel dynamic model-based reinforcement learning environment in Ray architecture.

Marine Autonomy Intern – Remote *Jul 2024 – Aug 2024*
Zeabuz *Trondheim, Norway*

- Worked as a remote intern at marine autonomy company Zeabuz under supervision of Dr Emil Hjelseth Thyri.
- Developed a deep learning model for predicting the trajectory of surface vessels in a region based on historical AIS data available for vessels operating at Stockholm area in Sweden.
- The model is intended to be integrated with the motion planner of the autonomous passenger ferry operating in Stockholm.

Engineering Intern *Dec 2022 – Feb 2023*
Dimension Renewables Pvt. Ltd. *India*

- Demonstrated model of Active Heave Compensation system on a ship.
- Developed control system code for the system.
- Made CAD model for the demonstration model of system.
- Used ROS 1 network for the system architecture development.

KEY PROJECTS

ARITRA VRX 2023 – Team Member *Aug 2023 – Nov 2023*
Participated and won 1st position globally in Virtual Robot X Competition 2023 organised by RoboNation, USA. Worked on developing solutions for VRX 2023 competition by RoboNation including Station Keeping, Waypoint tracking, Perception, Acoustic perception, Acoustic pinging tracking, Scan, Dock and Deliver along with team members. Developed dockerised solution for the competition in ROS 2.

ARITRA ASV – Project Member *Sep 2022 – Aug 2023*
Participated and won 3rd position in Njord Autonomous Ship Challenge 2023 Competition organised by Norwegian University of Science and Technology (NTNU), Trondheim, Norway. Team mentored by Dr Abhilash Somayajula and Dr Atmanand MA, former director of National Institute of Ocean Technology (NIOT), Ministry of Earth Sciences, Govt of India . Developed custom extended Kalman filter code for fusion of GPS and IMU sensors in the vessel. Implemented Line of Sight guidance law with PD control

for station keeping and navigation. Also developed YOLOv8 detection algorithm for buoy detection and COLREG compliant Artificial Potential Field (APF) algorithm for obstacle avoidance along with team members.

UAV Delivery & Aerial Robotics and Aeromodelling Club, IIT Madras *Jul 2022 – Mar 2023*
Worked on and delivered UAV for delivering medicines around the IIT Madras campus. Developed code for the autonomous navigation of the drone. Integration of 4G LTE with the drone and retrieving website data with ROS network. Setup communication via MQTT protocol to drone and control via mobile application. Setup indoor localisation with UWB technology for UAV localisation.

Scene Reconstruction using Neural Radiance Fields (NeRFs)
Scene reconstruction using NeRFs in SeaThru NeRF underwater image dataset using NeRF studio framework as part of course "Machine Learning for Ocean Engineers" under guidance of Dr Abhilash Somayajula.

Motion Planning for Moving Targets using Kuka Youbot Manipulator
Motion planning algorithm for catching moving targets using Kuka Youbot mobile manipulator as part of course "Introduction to Motion Planning" under guidance of Dr Bijo Sebastian.

POSITIONS OF RESPONSIBILITY

Team Lead, ARITRA – MAVLAB, IIT Madras *Dec 2022 – May 2025*
Managing the marine autonomous surface vessel competition team Aritra with special focus on sensor integration, sensor fusion and embedded systems module. Organising and preparing for lake test events at MAVLAB. Organising various events and pitching research work to companies and visiting groups in MAVLAB and also chairing MAVLAB Seminar series along with colleagues.

Exhibitions Coordinator, Shows and Exhibitions Vertical *Jul 2022 – Jan 2023*
Served as a coordinator in exhibitions vertical of SHASTRA, Asia's largest student-run college festival. Contacted several outside companies and brought them to IIT Madras campus for exhibition. Conducted Institute Open House events, in which we brought around 700 school children from outside to visit the research lab in IIT Madras campus. Organised SHASTRA Juniors event for providing an opportunity to school students to exhibit their products in IIT Madras campus.

HONOURS AND AWARDS

- **3rd place** – Njord Autonomous Ship Challenge 2023 (global), awarded a grant of 10,000 NOK along with team. Honoured by the Ambassador of Norway to India, Ms May-Elin Stener and Deputy Foreign Minister of Norway, Mr Andreas Motzfeldt Kravik.
- **1st place** – Virtual Robot X Competition 2023; won a grant of 1,500 USD and honoured by Mr V Kamakoti, Director of IIT Madras, as part of team Aritra.
- **Featured** in IEEE OES Beacon December 2023 edition as part of team Aritra and Njord autonomous ship competition 2023.
- **Selected** for IEEE Oceans Halifax 2024 as a student poster competition participant and secured a grant of 3,000 CAD jointly by Office of Naval Research, Marine Technology Society and IEEE OES.
- **Selected** for Underwater Technology Conference 2025 in Taiwan as a fully funded student poster competition participant.
- **Secured** a research initiation grant at Centre for Robotics and Autonomous Systems, INESC TEC, Porto, Portugal for working on an AUV project under Professor Nuno Alexandre Cruz.